Course Management System

Project Documentation

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1. Introduction

The Course Management System (CMS) is a web-based application developed to simplify the process of managing academic courses, users, and student registrations. The system supports three distinct roles—Admin, Teacher, and Student—each with clearly defined responsibilities. The project is implemented using Java Servlet and JSP technology with MySQL as the backend database.

2. Objectives of the Project

The main objectives of this project are:

- - To provide a centralized platform for course management.
- - To enable role-based access and operations.
- - To allow students to register for available courses.
- - To ensure secure session-based authentication.

3. System Overview

The CMS follows a typical MVC-style architecture where JSP pages act as the view layer, Servlets handle business logic, and DAO classes manage database interactions. User authentication determines access to role-specific dashboards.

4. Functional Description

Admin Module: The admin is responsible for creating courses and assigning teachers to those courses. Admins can view all existing courses and manage system-level operations.

Teacher Module: Teachers can view courses assigned to them and see the list of students registered in each course.

Student Module: Students can browse available courses and register themselves for desired courses. They can also view their registered course list.

5. Technologies Used

- - Programming Language: Java
- - Web Technologies: JSP, Servlets
- - Database: MySQL
- - Server: Apache Tomcat
- - IDE: Eclipse

6. Database Design

The database is implemented using MySQL and consists of three core tables: **users, courses,** and **registrations**. The users table stores authentication and role information. The courses table stores course details along with assigned teacher IDs. The registrations table maintains a many-to-many relationship between students and courses.

7. Setup and Installation

To set up the project, follow these steps:

- - Install MySQL and create a database named cmsdb.
- - Import the provided cms.sql file into the database.
- - Configure database credentials in DBUtil.java.
- - Import the project into an IDE as a Maven project.
- - Deploy the project on Apache Tomcat server.

8. Security and Session Management

The application uses HTTP sessions to manage logged-in users. Upon successful login, user information is stored in the session. Logout invalidates the session to prevent unauthorized access.

9. Conclusion

The Course Management System provides an efficient and organized solution for managing courses and student registrations. The modular architecture allows future enhancements such as password hashing, advanced authorization, and UI improvements.